

# Shell matching fixes the trace of transported flags

MathlibAnnex.CStarAlgebra.CAR.trace\_transportedFlag

## theorem

Recovers exact trace values from the initial and final supports of shell links.

## Statement

For the completed CAR algebra with normalized trace  $\tau$  and a fixed shell family, the transported projection  $f_{j,n} = \alpha_j(f_n)$  satisfies  $\tau(f_{j,n}) = 2^{-n}$  for every index  $j$  and every  $n \geq 0$ . Here  $f_n$  is the root projection at the matrix stage of size  $2^n$ .

## Assumptions

Let  $A$  be the completed CAR algebra with matrix stages  $M_{2^n}(\mathbb{C})$  embedded by  $a \mapsto a \otimes I_2$ . Write  $f_n$  for the image of the first diagonal matrix unit,  $p_n = f_n - f_{n+1}$ , and  $\omega$  for the root state characterized by  $\omega(\iota_n(a)) = a_{00}$  on each stage, where  $\iota_n$  is its canonical embedding.

Index chosen pure-state Gelfand-Naimark-Segal (GNS) representations by their unitary-equivalence classes  $j \in I$ , with root index  $o$  and representative states  $\varphi_j$ .

A fixed shell family consists of complex-linear unital star automorphisms  $\alpha_j$  and elements  $w_{j,n} \in A$  satisfying  $\varphi_j(\alpha_j(a)) = \omega(a)$ ,  $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$  and  $w_{j,n} w_{j,n}^* = p_n$ .

At the root,  $\alpha_o$  is the identity and  $w_{o,n} = p_n$ .

The functional  $\tau$  is the normalized CAR trace defined in the trace construction. In particular  $\tau(f_n) = 2^{-n}$ . The same family and the same automorphism  $\alpha_j$  are used for all values of  $n$ .

## Conclusion

Transported flags and root flags have the same trace at each level. This supplies the decaying scalar values needed to estimate their action on vectors implementing the trace.

No new automorphism is chosen and trace-invariance of arbitrary automorphisms is not assumed. The proof uses the exact support identities of the given links and finite induction; it does not infer that the projections themselves converge in operator norm.

## Proof route

For each shell link, traciality gives  $\tau(w_{j,n}^* w_{j,n}) = \tau(w_{j,n} w_{j,n}^*)$ . Its support identities turn this into  $\tau(f_{j,n}) - \tau(f_{j,n+1}) = \tau(f_n) - \tau(f_{n+1})$ . Both flags start at 1, so induction forces equal trace values at every level. The root value is the normalized trace of a rank-one diagonal matrix projection.

### ▼ Proof steps

1. Use  $w_{j,n}^* w_{j,n} = f_{j,n} - f_{j,n+1}$  and  $w_{j,n} w_{j,n}^* = f_n - f_{n+1}$  in the trace identity.
2. At  $n = 0$  both flags are 1. If their traces agree at  $n$ , equality of the successive differences gives agreement at  $n + 1$ .
3. At stage  $n$ , the root projection has exactly one diagonal entry equal to 1. Dividing its matrix trace by  $2^n$  gives the stated value.

## Main citations

- Exact statement and proof
- A fixed family of compatible shell data
- Automorphisms and partial-isometry shell links
- The root flag transported by the chosen automorphism
- The transported flag starts at the unit
- Initial support of a shell link
- Final support of a shell link
- The trace identity
- Trace of a root projection
- Shell matching preserves each flag trace
- The continuous trace on the completion

## Lean source signature (exact)

```
@[simp]
theorem trace_transportedFlag (family : RepresentativeShellFamily)
  (i : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit) (n : ℕ) :
  trace (transportedFlag family i n) = (2 ^ n : ℂ)-1
```

| In the source                      | Mathematical meaning                                                                                                                             |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| family : RepresentativeShellFamily | The fixed shell family $(\alpha_j, w_{j,n})$ with the support and root identities in the assumptions, in the completed CAR algebra $A$ .         |
| i : GNSClass Limit; n : ℕ          | The source index $i$ is the pure-GNS class $j$ of the statement, and $n \geq 0$ is the matrix stage. The same $\alpha_j$ is used for every $n$ . |
| transportedFlag family i n         | The projection $f_{j,n} = \alpha_j(f_n)$ , where $f_n$ is the first diagonal matrix projection at stage $M_{2^n}(\mathbb{C})$ .                  |

| In the source                                   | Mathematical meaning                                                                                                                                             |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>trace (transportedFlag family i n)</code> | The normalized CAR trace $\tau$ evaluated on that projection:<br>$\tau(f_{j,n})$ .                                                                               |
| <code>(2 ^ n : ℂ)<sup>-1</sup></code>           | The complex scalar $2^{-n}$ , the reciprocal of $2^n$ . Thus the entire conclusion is $\tau(f_{j,n}) = 2^{-n}$ ; it is not a norm estimate or a different trace. |

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▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.trace_transportFlag`

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